MRI-based inverse finite element characterization of ascending aortic aneurysms using Bayesian optimization

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Introduction

Clinical assessment of ascending aortic aneurysms is mainly based on geometric criteria such as maximum aortic diameter, which may not adequately reflect the patient-specific mechanical state of the arterial wall. This study aimed to develop a preliminary computational framework for the *in vivo* mechanical characterization of ascending aortic aneurysms by combining magnetic resonance imaging (MRI), inverse finite element analysis and Bayesian optimization.

Methods

Patient-specific aortic geometries were reconstructed from MRI acquisitions at the diastolic and the systolic phases. The segmented luminal surfaces were smoothed and processed to generate three-dimensional aortic wall models with a uniform thickness of 2.00 mm. An approximate zero-pressure configuration was estimated using a pull-back procedure. Finite element models were developed in Abaqus using the anisotropic Gasser–Ogden–Holzapfel constitutive model. The transition from diastole to systole was computed by applying an internal pressure increment and MRI-derived displacements. External tissue support was represented using regional elastic foundation conditions. For each simulation, finite element displacements were compared with MRI-derived displacement fields using the normalized root mean square error. Gaussian process regression and Bayesian optimization were subsequently used to identify plausible regions of the material parameter space. The identified parameter sets were independently assessed against patient-specific ex vivo biaxial stress–stretch curves.

Results

The proposed methodology was initially evaluated in a reference patient and subsequently explored in three additional cases. For the reference patient, repeated Bayesian optimization runs produced consistent solutions and generated theoretical stress–stretch curves showing reasonable agreement with ex vivo measurements.

In patients exhibiting lower displacement magnitudes, identifiability decreased because several material combinations produced similar kinematic responses. The use of vectorial displacement components instead of displacement magnitude improved discrimination in one of these cases. Conversely, displacement fields potentially affected by rigid-body motion or limited volumetric variation were insufficient for reliable material characterization. A preliminary heterogeneous model distinguishing the aneurysmal region from the remaining wall improved the spatial coherence of the simulated displacement field, although the increased number of parameters reduced identifiability.

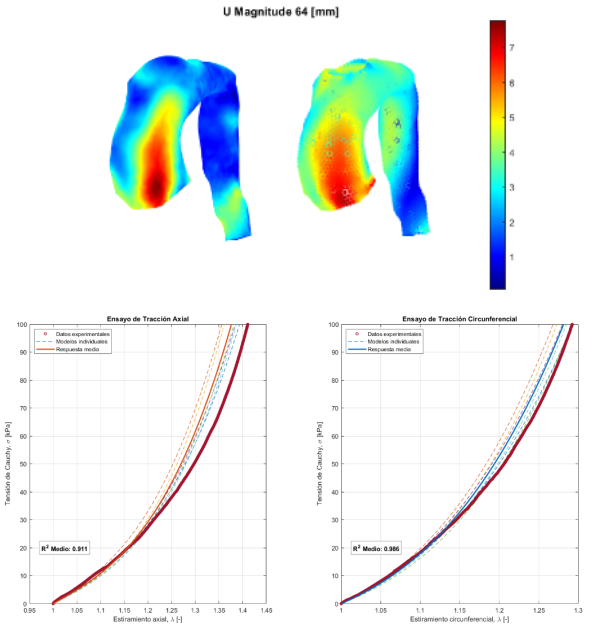


Figure 1: Comparison of the MRI-derived displacement (in mm) field (left) and the predicted field using the characterized material (right), together with axial and circumferential cauchy stress–stretch curves. Thick curves show the experimental data, dashed lines the individual model fits, and the solid line their mean model response.

Conclusions

MRI-derived displacement fields can provide useful information for the inverse characterization of ascending aortic wall properties when they represent effective tissue deformation. Regional displacement metrics, vectorial error formulations, and ex vivo validation improve the biomechanical interpretation of the identified solutions (Figure 1). Although further validation in larger patient cohorts is required, the proposed framework represents a promising basis for patient-specific *in vivo* characterization of ascending aortic aneurysms.

References

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